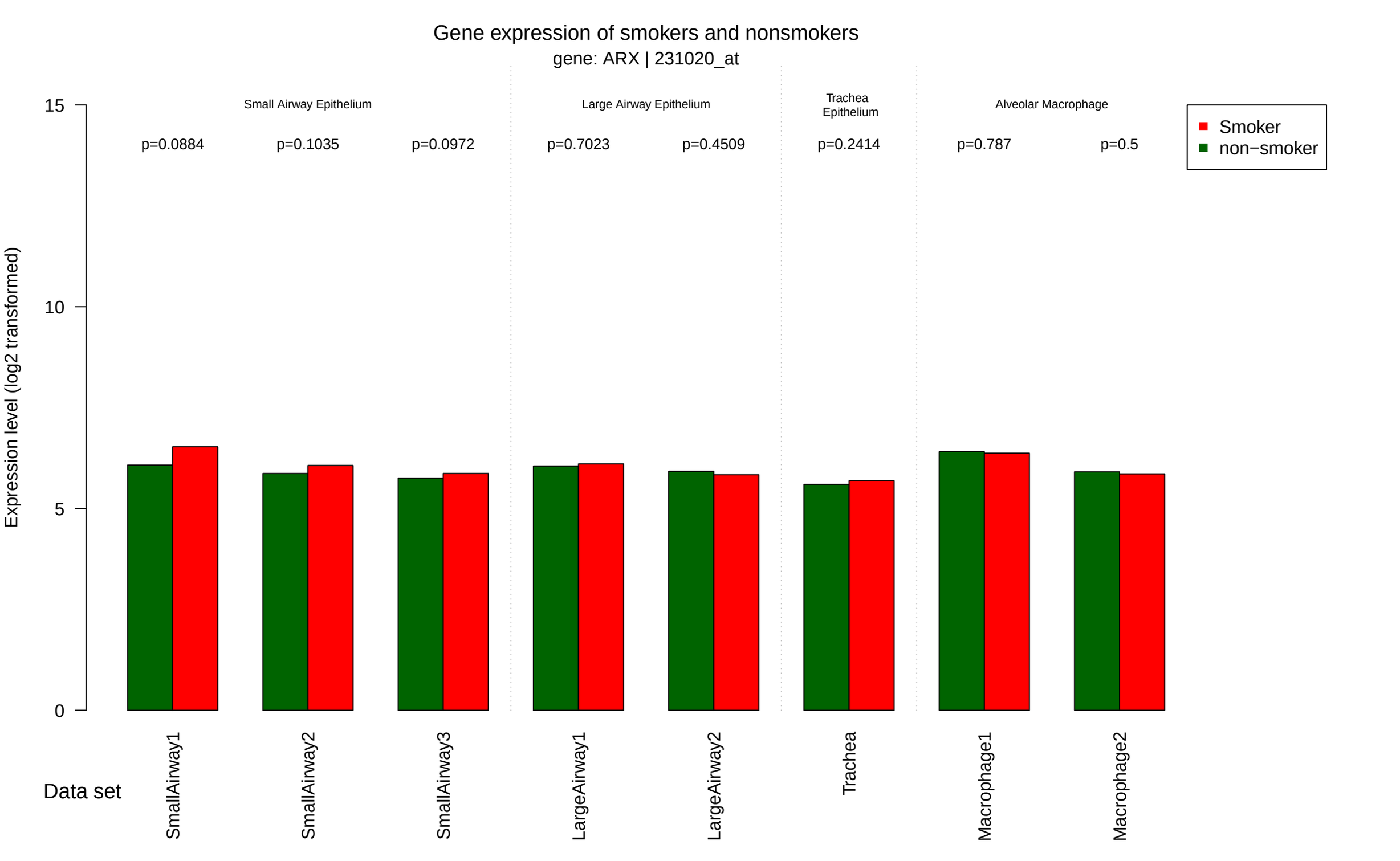
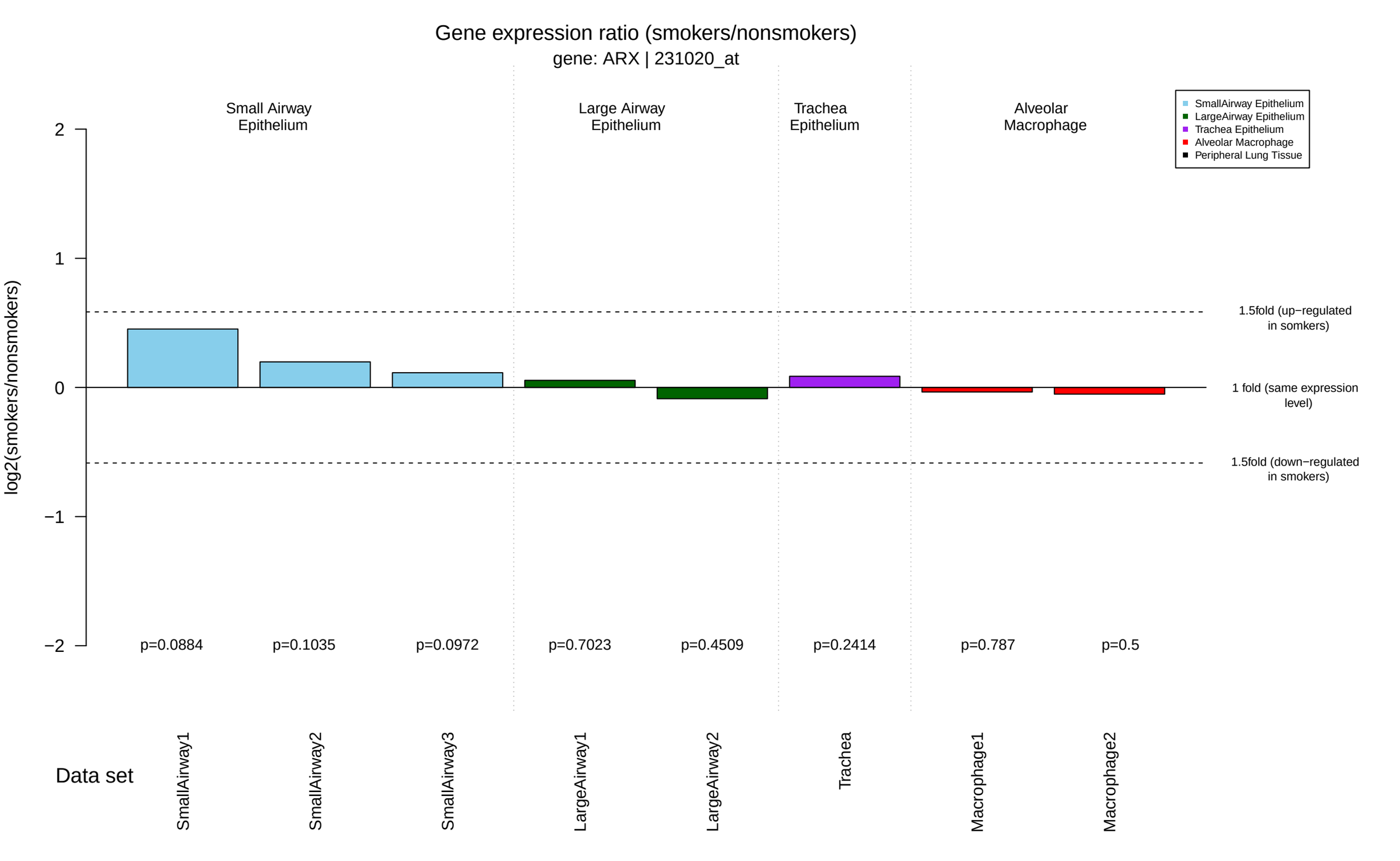
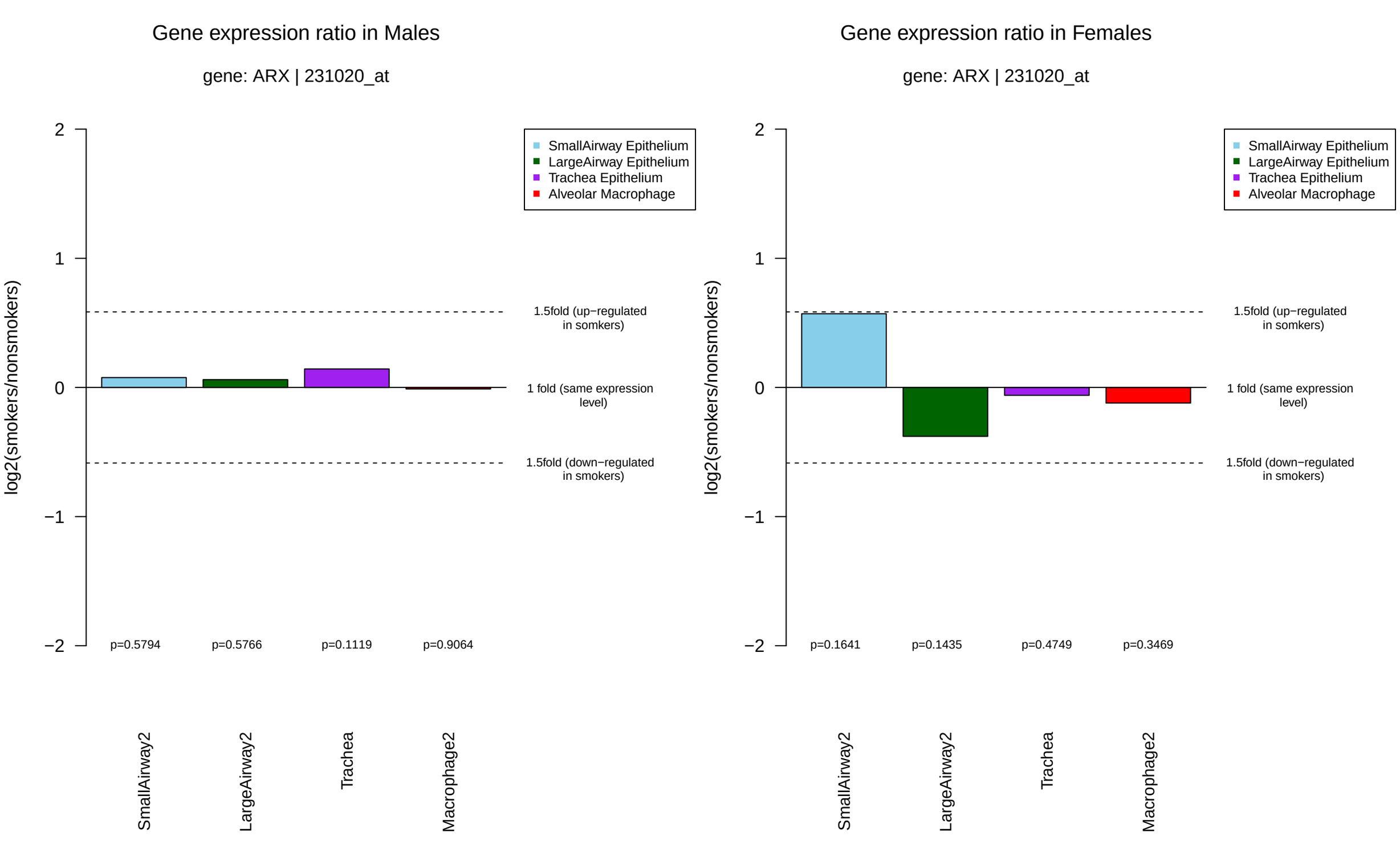
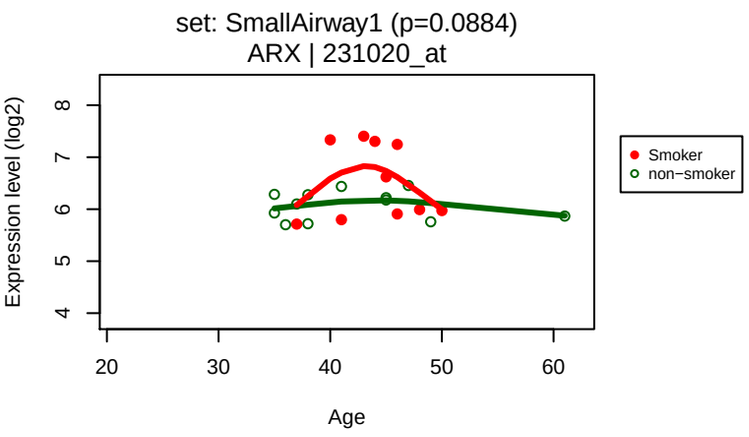
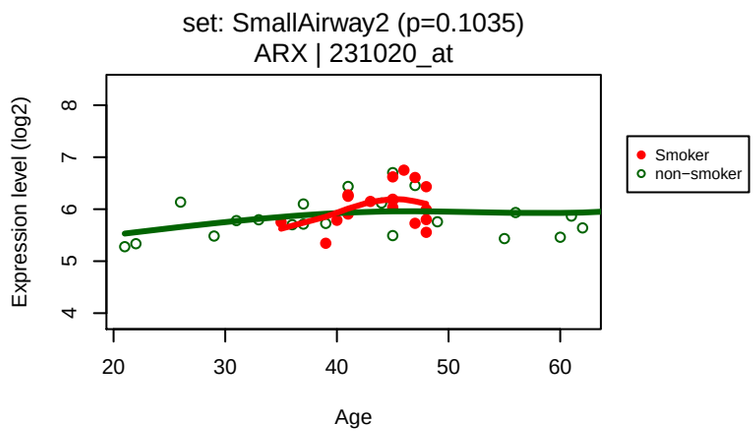
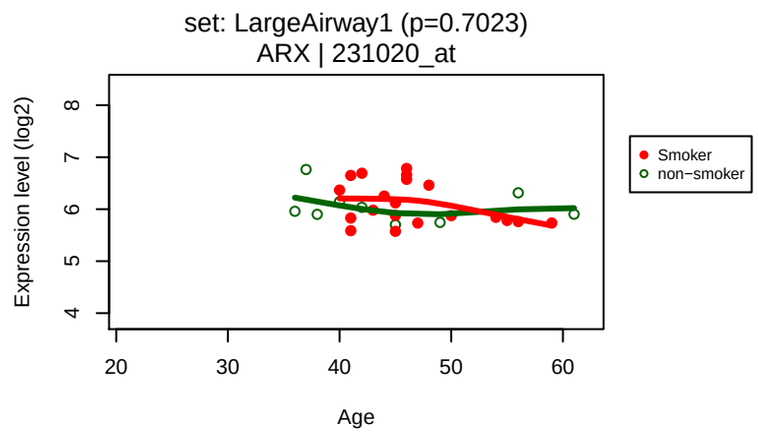

Gene expression of smokers and nonsmokers
gene: ARX | 231020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0884
p=0.1035
p=0.0972
p=0.7023
p=0.4509
p=0.2414
p=0.787
p=0.5
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

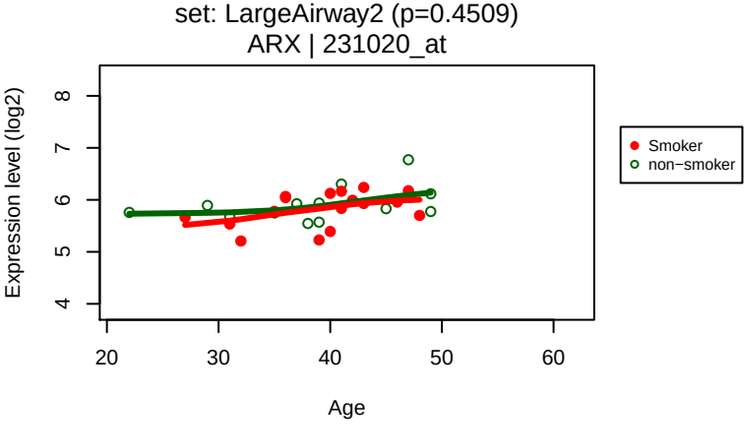

Gene expression ratio (smokers/nonsmokers)
gene: ARX | 231020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0884
p=0.1035
p=0.0972
p=0.7023
p=0.4509
p=0.2414
p=0.787
p=0.5
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

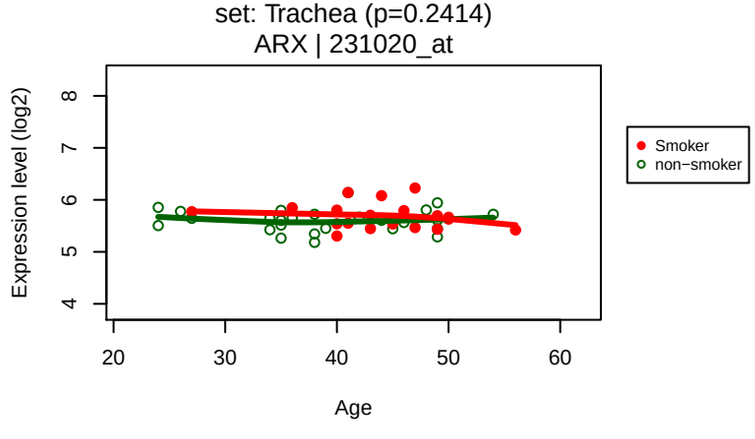

Gene expression ratio in Males
gene: ARX | 231020_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5794
p=0.5766
p=0.1119
p=0.9064
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARX | 231020_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1641
p=0.1435
p=0.4749
p=0.3469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

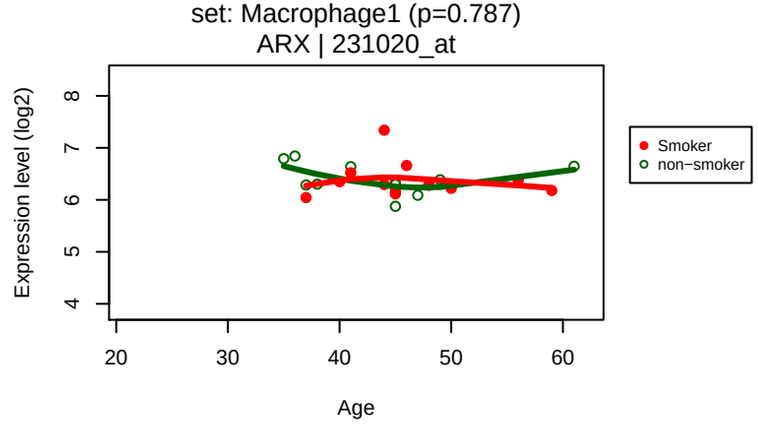

set: SmallAirway1 (p=0.0884)
ARX | 231020_at
Expression level (log2)
Age
Smoker
non−smoker

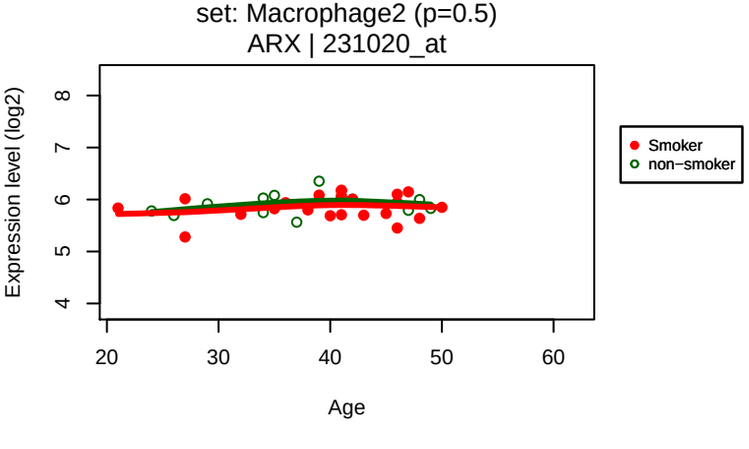

set: SmallAirway2 (p=0.1035)
ARX | 231020_at
Expression level (log2)
Age
Smoker
non−smoker

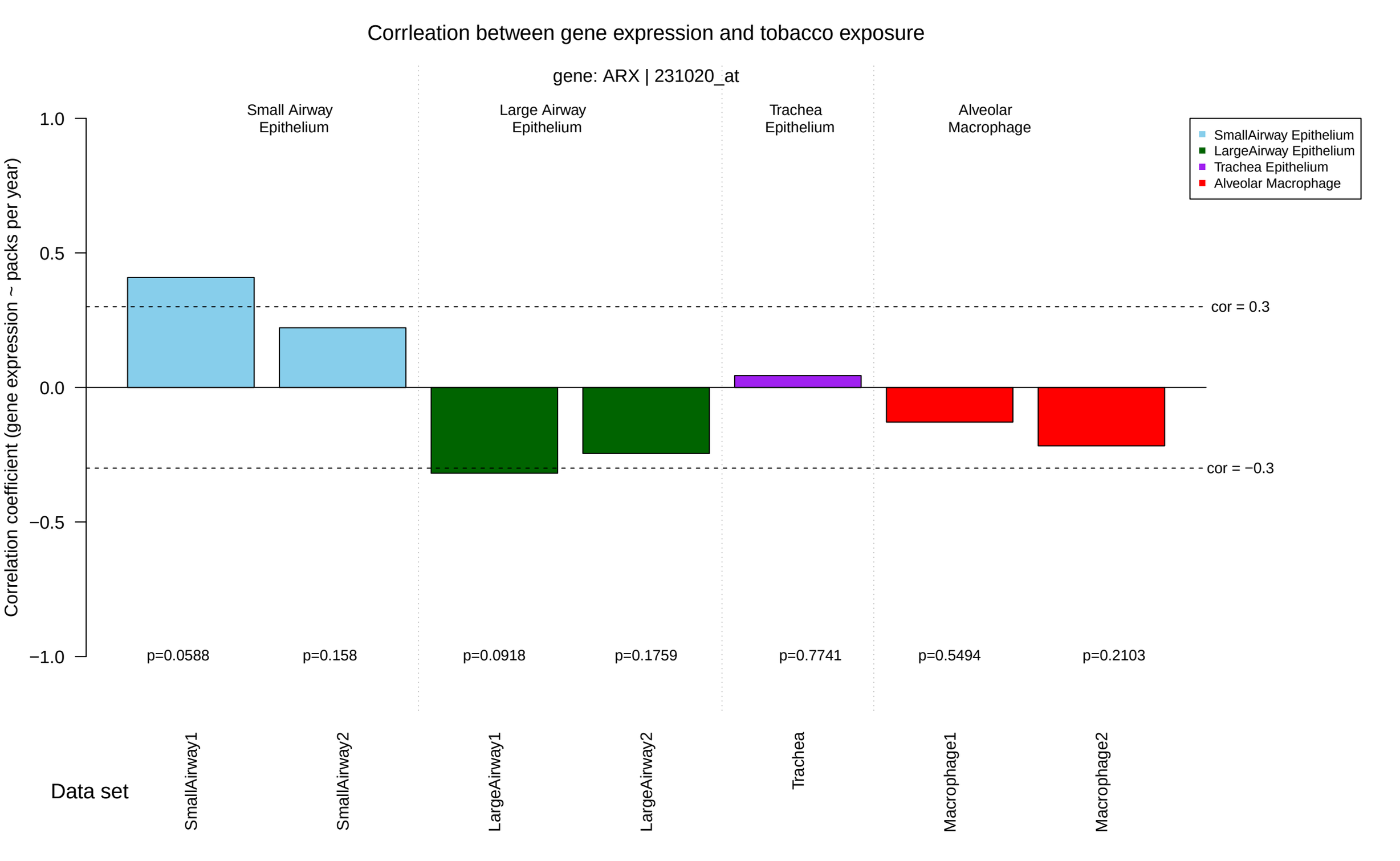

set: LargeAirway1 (p=0.7023)
ARX | 231020_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4509)
ARX | 231020_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2414)
ARX | 231020_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.787)
ARX | 231020_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5)
ARX | 231020_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARX | 231020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0588
p=0.158
p=0.0918
p=0.1759
p=0.7741
p=0.5494
p=0.2103
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2